

Model Questions for 2025 Undergraduate Mathematics Olympiad

Algebra and Complex Numbers

1. If z is a complex number such that $|z| = 1$, find the value of

$$\left(\frac{1+z}{1+\bar{z}} \right)^2.$$

Solution: Since $|z| = 1$, we have $z\bar{z} = 1$ or $\bar{z} = \frac{1}{z}$.

$$\frac{1+z}{1+\bar{z}} = \frac{1+z}{1+\frac{1}{z}} = \frac{1+z}{\frac{z+1}{z}} = z.$$

Therefore,

$$\left(\frac{1+z}{1+\bar{z}} \right)^2 = z^2.$$

2. Find all pairs (a, b) of real numbers for which there exists a unique symmetric 2×2 matrix M with $\text{trace}(M) = a$ and $\det(M) = b$.

Solution: Let $M = \begin{bmatrix} x & y \\ y & z \end{bmatrix}$. Then $\text{trace}(M) = x+z = a$ and $\det(M) = xz - y^2 = b$.

We have $z = a - x$, so $\det(M) = x(a - x) - y^2 = b$.

For uniqueness, the discriminant must be zero:

$$\Delta_x = a^2 - 4(y^2 + b) = 0 \Rightarrow y^2 = \frac{a^2}{4} - b.$$

For unique $y = 0$, we require $\frac{a^2}{4} - b = 0$, i.e., $a^2 = 4b$.

3. When $P(x)$ is divided by $x-1$, remainder is 1, and when divided by $x+1$, remainder is -1. Find remainder when $P(x)$ is divided by $x^2 - 1$.

Solution: Let remainder be $ax + b$. Then:

$$P(1) = a + b = 1, \quad P(-1) = -a + b = -1.$$

Solving: $a = 1, b = 0$. So remainder is x .

4. If $x, y, z > 0$, prove that

$$\frac{2}{x+y} + \frac{2}{y+z} + \frac{2}{z+x} \geq \frac{9}{x+y+z}.$$

Solution: By Cauchy-Schwarz:

$$[(x+y) + (y+z) + (z+x)] \left(\frac{1}{x+y} + \frac{1}{y+z} + \frac{1}{z+x} \right) \geq 9.$$

So $2(x+y+z) \left(\frac{1}{x+y} + \frac{1}{y+z} + \frac{1}{z+x} \right) \geq 9$, hence the result.

Calculus

5. Let $f(x) = \sin x - \cos x$. Determine monotonicity and concavity in $[-\pi, \pi]$.

Solution: $f'(x) = \cos x + \sin x = \sqrt{2} \sin(x + \pi/4)$.

Critical points: $x = -\pi/4, 3\pi/4$.

$f''(x) = -\sin x + \cos x = \sqrt{2} \cos(x + \pi/4)$.

Inflection points: $x = -3\pi/4, \pi/4$.

6. Find k such that

$$f(x) = \begin{cases} kx^2, & x \leq 2 \\ 2x + k, & x > 2 \end{cases}$$

is continuous.

Solution: At $x = 2$: $4k = 4 + k \Rightarrow 3k = 4 \Rightarrow k = \frac{4}{3}$.

7. Find area enclosed by $y^2 = x$ and $y = x - 2$.

Solution: Intersection: $y^2 = y + 2 \Rightarrow y = -1, 2$. Points: $(1, -1), (4, 2)$.

$$\text{Area} = \int_{-1}^2 [(y+2) - y^2] dy = \left[\frac{y^2}{2} + 2y - \frac{y^3}{3} \right]_{-1}^2 = \frac{9}{2}.$$

8. Solve: $\frac{dP}{dt} = \frac{1}{2}P - \frac{1}{8}P^2$, with $P(0) = 100$.

Solution: Separate variables:

$$\frac{dP}{P(4-P)} = \frac{1}{8} dt.$$

Partial fractions and integrate gives:

$$P(t) = \frac{100e^{t/2}}{25e^{t/2} - 24}.$$

9. For $T(x, y) = 20 - x^2 - y^2$, find unit vector of most rapid decrease at $(-1, -3)$.

Solution: Gradient: $\nabla T = (-2x, -2y) = (2, 6)$ at $(-1, -3)$.

Direction: $(-2, -6)$. Unit vector: $\frac{(-1, -3)}{\sqrt{10}}$.

Rate: $-|\nabla T| = -2\sqrt{10}$.

Linear Algebra

10. Determine if $T(x, y, z) = (x - z, y - z, x - y, x + y + z)$ is one-to-one/onto.

Solution: Matrix representation shows kernel is trivial, so one-to-one.

Domain dimension 3, codomain dimension 4, so not onto.

11. Find characteristic polynomial and inverse of $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{pmatrix}$.

Solution: Characteristic polynomial: $\lambda^3 - 14\lambda^2 + 46\lambda + 1 = 0$.

$$\det A = -1, \text{ so } A^{-1} = \begin{pmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{pmatrix}.$$

12. Find basis for span of polynomials $p_1 = 1 - 2x + 3x^3$, $p_2 = 2 - 5x - 3x^2 + 6x^3$, $p_3 = x + 3x^2$, $p_4 = 5 - 8x + x^2 + 2x^3$, $p_5 = 2 - x + 4x^2 - 7x^3$.

Solution: Basis: $\{p_1, p_2, p_4\}$.

Relations: $p_3 = 2p_1 - p_2$, $p_5 = -p_1 - p_2 + p_4$.

13. Find kernel basis for $L(at^2 + bt + c) = (a + 2b)t + (b + c)$.

Solution: Kernel: $a = -2b, c = -b$. Basis: $\{-2t^2 + t - 1\}$.

Differential Equations

14. Solve: $y'' - 4y' + 4y = 0$, $y(0) = 0$, $y'(0) = 3$.

Solution: Characteristic: $(r - 2)^2 = 0$. Solution: $y = 3te^{2t}$.

15. Is $\mathbf{F} = \frac{c}{r^3}(x, y, z)$ solenoidal?

Solution: Yes, since $\nabla \cdot \left(\frac{\mathbf{r}}{r^3}\right) = 0$ for $r > 0$.

16. Work done by $\mathbf{F} = (x^4 - y^3, \cos y + x^3)$ around unit circle.

Solution: By Green's theorem:

$$W = \iint_D 3(x^2 + y^2) dA = 3\pi.$$

17. Volume of sphere using spherical coordinates.

Solution:

$$V = \int_0^{2\pi} \int_0^\pi \int_0^1 \rho^2 \sin \phi d\rho d\phi d\theta = \frac{4\pi}{3}.$$

18. Work by $\mathbf{F} = (x - y, y - z, z - x)$ around triangle in first octant.

Solution: Curl $\mathbf{F} = (1, 1, 1)$. By Stokes:

$$W = \sqrt{3} \times \text{Area} = \sqrt{3} \times \frac{\sqrt{3}}{2} = \frac{3}{2}.$$

Series and Sequences

19. If $x_n^2 - x_{n-1}x_{n+1} = 1$, show $x_{n+1} = ax_n - x_{n-1}$ for some constant a .

Solution: From given, $x_{n+1} = \frac{x_n^2 - 1}{x_{n-1}}$. Show that $a = \frac{x_n^2 + x_{n-1}^2 - 1}{x_n x_{n-1}}$ is constant.

20. Test convergence of $\sum \frac{n + \sqrt{n}}{n^2 - n}$.

Solution: Terms $\sim \frac{1}{n}$, so diverges by comparison with harmonic series.

21. Test convergence of $\sum \frac{(n!)^2}{(2n)!} x^n$.

Solution: Ratio test gives radius of convergence $|x| < 4$.

22. Find partial sum of $\sum \frac{\sqrt{k+1} - \sqrt{k}}{\sqrt{k^2 + k}}$.

Solution: Term $= \frac{1}{\sqrt{k}} - \frac{1}{\sqrt{k+1}}$, so $S_n = 1 - \frac{1}{\sqrt{n+1}}$.

Miscellaneous

23. Prove $\int_0^a \frac{f(x)}{f(x) + f(a-x)} dx = \frac{a}{2}$.

Solution: Use substitution $x \rightarrow a - x$ and add.

24. Maximize area with cost constraint for fencing.

Solution: Dimensions: 800 ft $\times$ 600 ft.

25. Inflection point of cubic with three roots is their average.

Solution: For $f(x) = a(x-r)(x-s)(x-t)$, $f''(x) = 0$ at $x = \frac{r+s+t}{3}$.

26. Prove $H_{2^n} \geq 1 + \frac{n}{2}$.

Solution: Group terms and bound each group below by $1/2$.

27. Compute $(1+i)^{100}$.

Solution: $1+i = \sqrt{2}e^{i\pi/4}$, so $(1+i)^{100} = 2^{50}e^{i25\pi} = -2^{50}$.